the Professor's equation and figure out how to lift humanity off Earth.

How do these rules play out in *Interstellar*?

Messaging Murph



When falling into and through the tesseract, Cooper truly does travel backward relative to our brane's time, from the era when Murph is an old woman to the era when she is ten years old. He does this in the sense that, looking at Murph in the tesseract bedrooms, he sees her ten years old. And he can move forward and backward relative to our brane's time (the bedroom's time) in the sense that he can look at Murph at various bedroom times by choosing which bedroom to look into. This does not violate rule 1 because Cooper has not reentered our brane. He remains outside it, in the tesseract's three-dimensional channel, and he looks into Murph's bedroom via light that travels forward in time from Murph to him.

But just as Cooper can't reenter our brane in Murph's ten-year-old era, so he can't send light to her. That would violate rule 1. The light could bring her information from Cooper's personal past, which is her future; information from the era when she is an old woman—backward-in-time information from one location in our brane to another. So there must be some sort of one-way spacetime barrier between ten-year-old Murph in her bedroom and Cooper in the tesseract, rather like a one-way mirror or a black-hole horizon. Light can travel from Murph to Cooper but not from Cooper to Murph.

In my scientist's interpretation of *Interstellar*, the one-way barrier has a simple origin: Cooper, in the tesseract, is always in ten-year-old Murph's future. Light can travel toward the future from Murph to him. It can't travel to the past from him to Murph.